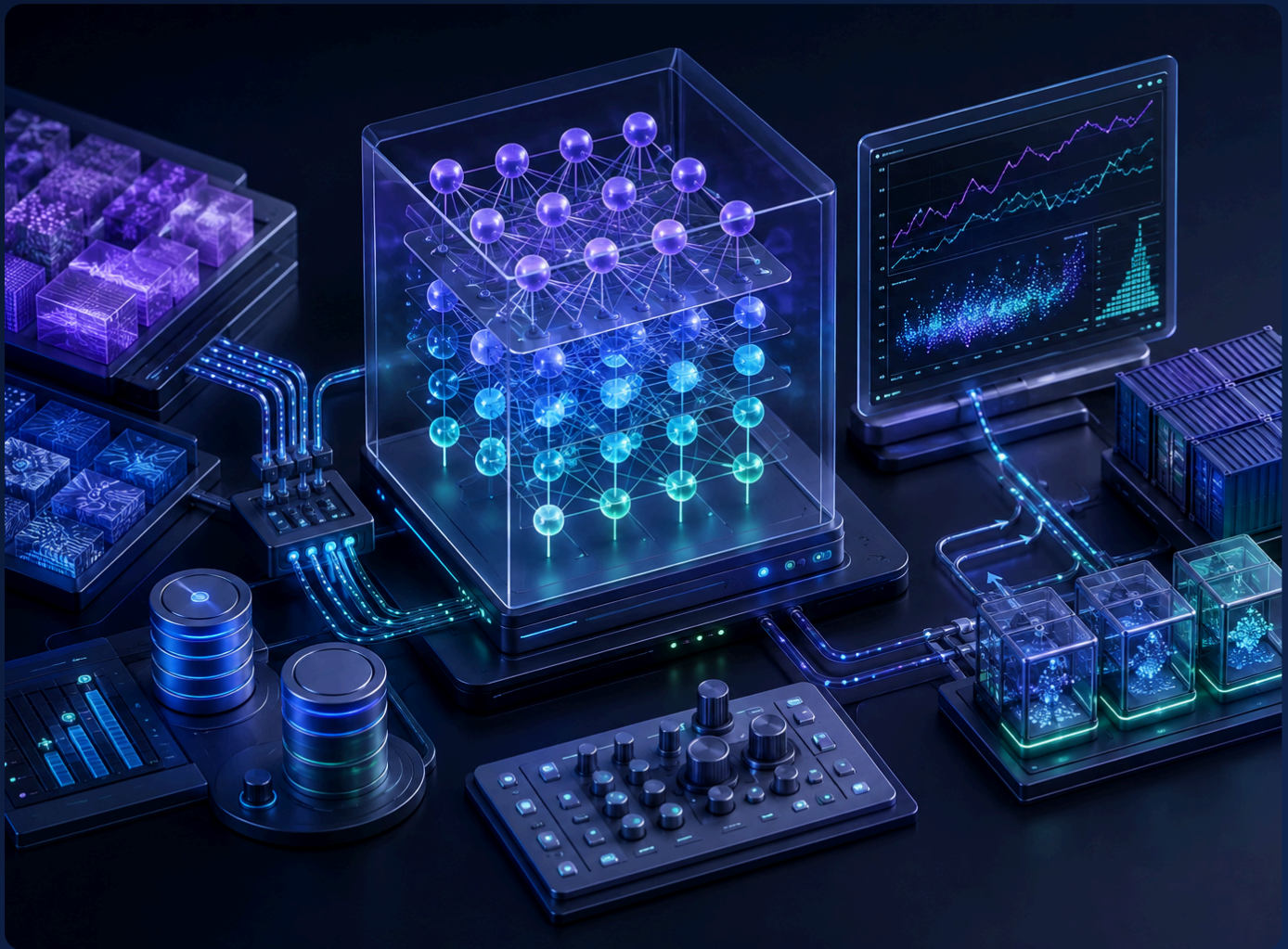


ARTIFICIAL INTELLIGENCE · MULTI-LEVEL

Applied Machine Learning Engineer Path

Build models. Test assumptions. Deliver useful ML.



Data → ML → Deep Learning → CV/NLP → LLM/SLM Engineering → MLOps.

🕒 20-28 weeks

📁 6 modules

🎓 Professional learning

Explore the learning experience →

Practical learning. Clear progression. Work you can explain.

YOUR NEXT CAPABILITY

What will you be able to do?

A production-oriented machine learning path spanning classical ML, deep learning, computer vision, NLP, voice, multimodal AI, model adaptation and MLOps.



Build and evaluate classical ML systems



Develop deep learning solutions for CV and NLP



Fine-tune models with PEFT, LoRA and QLoRA



Train a small educational transformer



Deploy and monitor models with MLOps



Who this is for

- AI and data students
- Data analysts
- Software developers entering ML
- Career switchers



Your starting point

- Basic programming
- Python and math bridges are included

Your progression at a glance

FROM CONCEPT TO EVIDENCE



Learn it. Apply it. Explain it.

Move from guided concepts to practical exercises and a coherent piece of work. Use the course resources to revisit difficult ideas and make your reasoning visible.

THE CURRICULUM

A clear route through the subject.

Every module builds towards practical work. The course outline below is aligned with the academy's current program catalog.

1 Data, Math and Python Bridge

Practical foundations for ML work.

- Python for data
- Statistics essentials
- Data preparation
- SQL and visualization

2 Classical Machine Learning

Supervised, unsupervised and evaluation workflows.

- Regression and classification
- Feature engineering
- Clustering
- Metrics and model selection

3 Deep Learning

Neural networks and practical training.

- Neural network foundations
- Optimization
- CNNs
- Sequence and transformer foundations

4 Computer Vision, NLP and Multimodal

Modern perception and language systems.

- Classification and detection
- Segmentation, tracking and OCR
- NLP and transformers
- Voice and multimodal workflows

5 LLM and SLM Engineering

Adapt, compress and understand language models.

- Dataset preparation and evaluation
- Supervised fine-tuning
- PEFT, LoRA and QLoRA
- Quantization and distillation

6 Educational Transformer and MLOps

Understand internals and operate models.

- Tokenizer and embeddings
- Transformer blocks and training loop
- Model serving
- Monitoring, drift and capstone

Practice with purpose

Prepare a dataset, establish a baseline, compare models and package a reproducible prediction workflow with documented limitations.

BUILD SOMETHING THAT DEMONSTRATES YOUR LEARNING

End-to-end prediction service

Prepare a dataset, establish a baseline, compare models and package a reproducible prediction workflow with documented limitations.

**Reproducible data pipeline****Model evaluation report****Prediction service and monitoring plan**

Tools and working methods

Python and Pandas

scikit-learn

PyTorch

OpenCV and YOLO concepts

Transformers and PEFT

Model serving and MLOps

Tools are part of the learning workflow, not partner endorsements. Examples may evolve while preserving the learning outcomes.



Delivery

Hybrid learning · 20–28 weeks

Confirm the current schedule, delivery arrangements and any prerequisites before enrolling.



Completion

A completion certificate is available after the required course work and a passing assessment.

Course completion is not a job, placement or funding guarantee.

Your next steps

- 1 Review the curriculum and your starting point.
- 2 Review current enrollment options and delivery details.
- 3 Start learning, keep your evidence and track your progress.

[Open the course experience →](#)

Customer-preview edition. Interactive links open the local academy prototype. Confirm course availability, schedule and commercial terms with the academy.

INSIDE THE LEARNING EXPERIENCE

Prevent leakage in a customer-risk model

A model predicts whether a customer will leave during the next month. The dataset includes activity timestamps and a field populated only after an account is closed.

Understand the concept

Define the prediction time first. A feature is valid only if it would be available at that time in real use. The post-closure field leaks the target and must not enter training. Split data in a way that reflects deployment, and fit learned preprocessing only on the training partition. Compare a simple baseline before choosing a more complex model.

Worked example

1. Fix a prediction cutoff for each example.
2. Exclude features created after that cutoff.
3. Separate train, validation and final test periods.
4. Fit preprocessing and the model on training data.
5. Choose the operating threshold on validation data.
6. Evaluate once on the held-out test set.

Interpret the result

A lower but credible held-out score is more useful than a misleading score caused by leakage. Report the confusion matrix and precision/recall at the chosen threshold, together with subgroup limitations and the cost of false decisions.

Knowledge check

Why should the final test set not be used repeatedly to choose model settings?

Explanation: Repeated choices adapt the model-selection process to the test set, weakening its value as an independent estimate.

YOUR PRACTICAL ASSIGNMENT

Build evidence, not just familiarity.

Prevent leakage in a customer-risk model



Your task

- Document when every candidate feature becomes available.
- Build a reproducible baseline pipeline and compare one stronger model.
- Prepare a model card with evaluation limits and monitoring signals.



Submission review criteria

- No target or temporal leakage
- Reproducible preprocessing and separate final evaluation
- Threshold choice explained in terms of the workflow

Submit

A working artifact or analysis, a short explanation of your decisions, and evidence that addresses each review criterion.

Reflect

Describe one failed approach, how you diagnosed it, and what you would test next. Identify limitations instead of hiding them.

Your project connection

Prepare a dataset, establish a baseline, compare models and package a reproducible prediction workflow with documented limitations.

Use the sample assignment as a stepping stone toward the course project, not as a substitute for the full curriculum.

[Explore the worked lesson](#) →

Proposed teaching sample for curriculum and UX review. Full lesson production, instructor review and delivery scheduling remain separate work.